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MILANO 1863**

**059686 - Finite Element Simulation for Mechanical Design
Mechanical Engineering
Automotive and Motorsport Engineering, M.Sc.**

FEM Exercise Summary

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1 Ex 1: FE model with Bar elements

1.1 Scope of the exercise

The scope of this exercise was to solve for the unknown displacements and reaction forces of a 2D triangular structure using the finite element method. The computations were carried out step by step in Excel and then numerically using Abaqus. The exercise's aim was also to study how the choice of element would influence the results of the simulations.

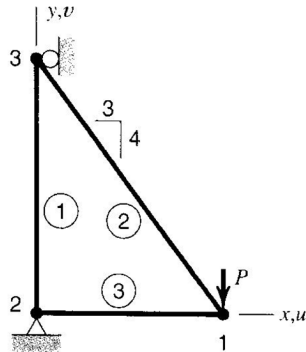
1.2 Model description

1.2.1 Geometry

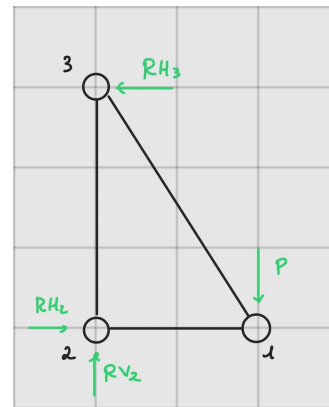
The triangular structure was modelled as a closed structure composed of three steel elements as shown in (Fig. 1a), the dimensions and other useful data are presented in (Tab. 1).

L_1	2000mm	P	10kN
L_2	2500mm	A	100mm ²
L_3	1500mm	E	206GPa

Table 1: Structure data



(a) Starting geometry



(b) Geometry modelled analytically

Figure 1: Geometry

1.2.2 Boundary conditions

The boundary conditions applied to the structure were a hinge placed in 2 and a roller moving along the vertical direction placed in 3, they can be seen in both (Fig. 1a) and (Fig. 1b). The structure is effectively a closed triangle and all the pieces are welded together with no other internal links.

1.2.3 Load

A vertical load was applied on 1 along the vertical direction, pointing downwards, of value $P = 10000N$.

1.2.4 Mesh

Truss elements were chosen to model the structure to study how the introduction of this element will affect the results. While using truss elements it's assumed that only axial forces will be transmitted between the elements in the whole structure, so effectively it's equivalent to introducing hinges at each node connection. Each side of the triangle was modelled using only one element, the nodes present in the structure were corresponding to the points 1, 2 and 3 in (Fig. 1b).

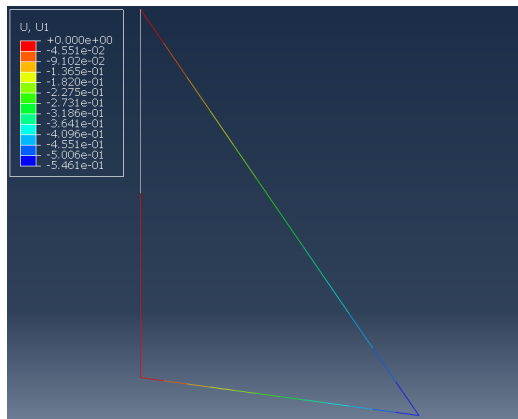
1.3 Simulation results

1.3.1 Displacements

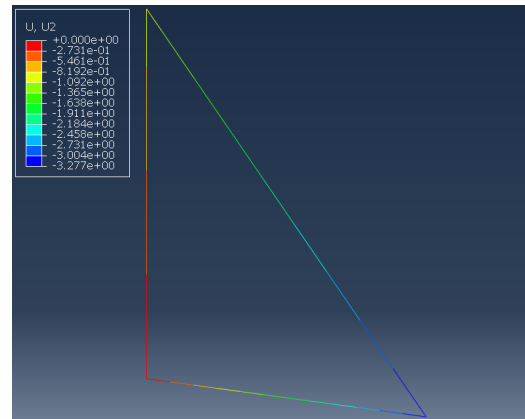
Analyzing the results coming from Abaqus is possible to see that the displacements shown are consistent with what one should expect. The horizontal and vertical displacement are maximum at the node where the load is applied, as shown in (Fig. 2a) and (Fig. 2b). In particular, the more interesting nodal displacements are the unknown ones, which are: u_1 , v_1 and v_3 . Their values are reported in (Tab. 2).

u_1	-0.5461 mm
v_1	-3.2767 mm
v_3	-0.9708 mm

Table 2: Abaqus displacement results



(a) Abaqus solution for horizontal displacement



(b) Abaqus solution for vertical displacement

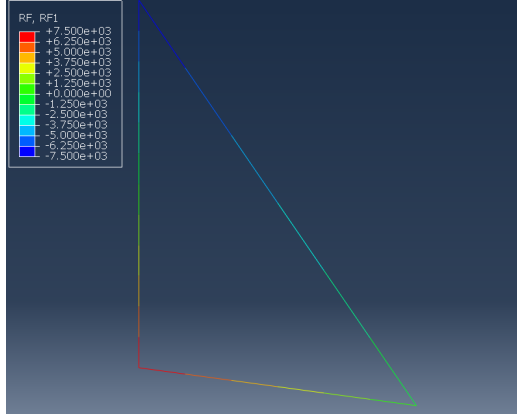
Figure 2: Resulting displacements

1.3.2 Reaction forces

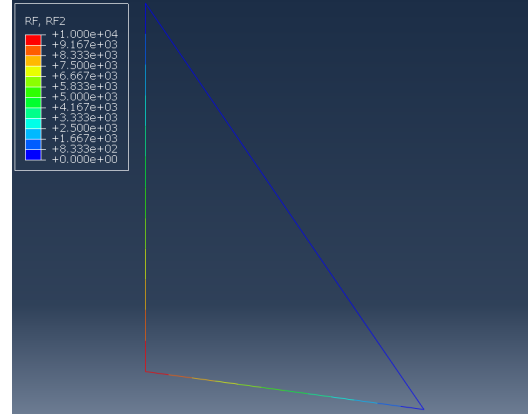
Also regarding the reaction forces is possible to see that the results are coherent with what one can expect from a structure like this. The reaction forces are maximum at the constrained nodes: in (Fig. 3a) where the horizontal reaction forces are shown it is clear that on node 2 is present a force directed towards the positive horizontal direction (right) due to the hinge while on node 3 an equal and opposite force is present, due to the presence of the roller. The same visual analysis can be carried out with (Fig. 3b) showing the vertical reaction force, which is only present in node 2 and is pointed upwards. The values are reported in (Tab. 3).

RV_2	10000 N
RH_2	7500 N
RH_3	-7500 N

Table 3: Abaqus reaction forces results



(a) Abaqus solution for horizontal reaction force



(b) Abaqus solution for vertical reaction force

Figure 3: Resulting reaction forces

1.4 Result analysis

In order to validate the results achieved with the Abaqus model two different approaches can be used to compare the results. A more analytical one computing the reaction forces through forces and moments equilibrium and one using the finite elements approach on Excel, solving each passage one at a time.

1.4.1 Analytical approach

The analytical approach (Fig. 1b) consists in converting the external constraints in their equivalent reaction forces applied on the structure and in placing in each node a hinge, this is done since the finite element used in the structure is a truss, which only transmits axial forces. Now is possible to compute the reaction forces through a vertical and horizontal force balance and a moment balance on node 2.

$$\begin{cases} RH_2 = RH_3 \\ RV_2 = P \\ RH_3 \cdot L_1 = P \cdot L_3 \end{cases} \longrightarrow \begin{cases} RH_2 = RH_3 \\ RV_2 = P \\ RH_3 = P \cdot \frac{L_3}{L_1} \end{cases} \longrightarrow \begin{cases} RH_2 = 7500 N \\ RV_2 = 10000 N \\ RH_3 = 7500 N \end{cases} \quad (1.1)$$

Comparing the analytical results obtained in (Eq. 1.1) with the results obtained with Abaqus in (Tab. 3), adjusting the signs for the different force orientations, it can be stated that the results of the reaction forces are coherent with each other since the values are exactly the same.

This coherence can be explained by the simplicity of the original structure and by the fact that the only load present on the structure is directly applied on a node of the FE model, this means that is not needed to approximate its effect with equivalent loads that can hinder the results.

1.4.2 FE model using Excel

Since it was a simple structure it was also possible to follow a more rigorous approach to the FE problem using Excel. The problem was solved step by step computing the stiffness matrix for each element and then combining them to create the structure's stiffness matrix $\underline{K}$. Then, rearranging the rows to highlight the free and constrained degrees of freedom, it was possible to compute the displacement of the free degrees of freedom $\underline{d}_F$ and the reaction forces $\underline{R}_C$, as shown in (Eq. 1.2).

$$\begin{cases} \underline{d}_F = \underline{K}_{FF}^{-1}(\underline{F}_F - \underline{K}_{FC}\underline{d}_C) \\ \underline{R}_C = \underline{K}_{CF}\underline{d}_F + \underline{K}_{CC}\underline{d}_C \end{cases} \quad (1.2)$$

Once again, the results found numerically with Abaqus were exactly equal, except for numerical rounding, to the results found rigorously with Excel. This should not come as a surprise since the system is geometrically very simple, only one concentrated load is applied and it's directly on a node.

2 Ex 2: FE model with Beam elements

2.1 Scope of the exercise

In this exercise the focus was to study the behaviour of beam elements and how to correctly implement them inside a FE model. In particular, it was shown how the element size, number and interconnections affect the results of a simulation. At first a simply supported beam was analyzed, then the triangular structure used in Ex.1 was adapted to the use of beam elements.

2.2 Model description: pin-pin beam

2.2.1 Geometry

The geometry of the model is very simple, consist in a 100 mm long steel beam with rectangular section of $[5 \times 10]\text{ mm}$, $E = 206000\text{ MPa}$ and $\nu = 0.33$.

2.2.2 Boundary conditions

Being a simply supported beam the boundary conditions are just two pins ad the edges of the beam.

2.2.3 Load

The load used in this study was a vertical distributed load along the beam of value 1 N/m in the negative direction.

2.2.4 Mesh

The element used in the study is a beam with cubic formulation, the number of elements used in the mesh varied to show different FE results.

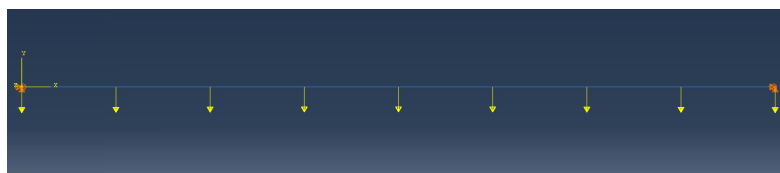


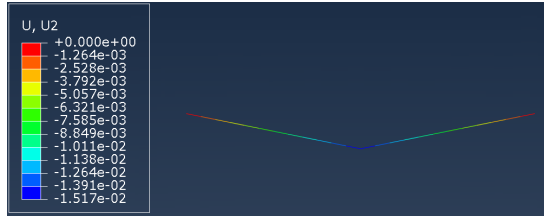
Figure 4: Loaded pin-pin beam in Abaqus

2.3 Model description: triangular structure

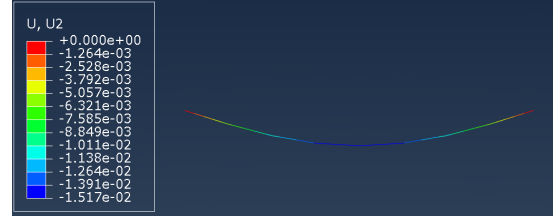
Since the aim of the simulation was to show the difference between beam and truss elements the geometry of the triangular structure is the same as the one in Ex.1. In one case the simulation was run with the exact same file by just changing the element type, while in a second moment some modifications were made to account for the different element formulation, in order to have comparable results. In particular, in this second case, the structure was not anymore entirely welded but three different parts were created and then connected by each other effectively creating hinge internal joints.

2.4 Simulation results: pin-pin beam

In (Fig. 5) are shown the vertical displacements, while in (Fig. 6) are shown the section moments present in the beam due to a distributed load, using a different number of element in each simulation.

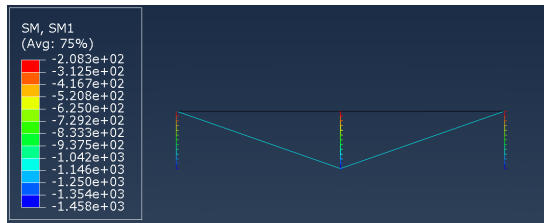


(a) Two elements simulation

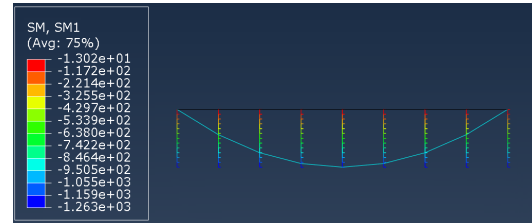


(b) Eight elements simulation

Figure 5: Beam vertical displacement with different elements number



(a) Two elements simulation

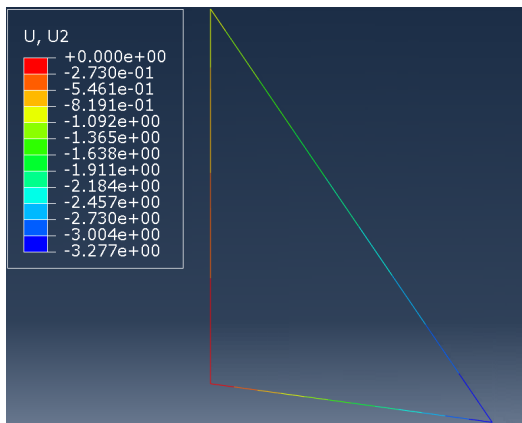


(b) Eight elements simulation

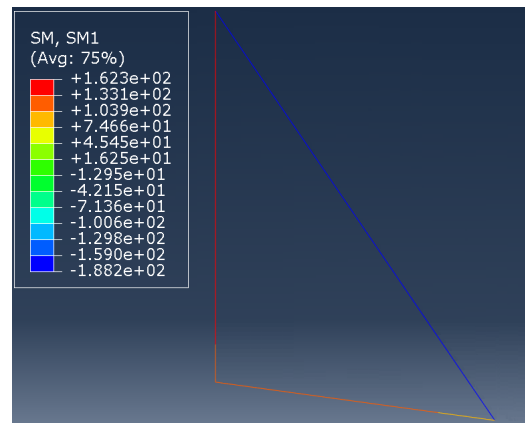
Figure 6: Beam section moment with different elements number

2.5 Simulation results: triangular structure

In (Fig. 7) are shown the results for the triangular structure using welded beams, the reaction forces are not present being the same present in (Fig. 3). While in (Fig. 8) are shown the same results but using beams connected internally by hinges.



(a) Vertical displacement



(b) Section moments

Figure 7: Triangular structure using welded beam elements

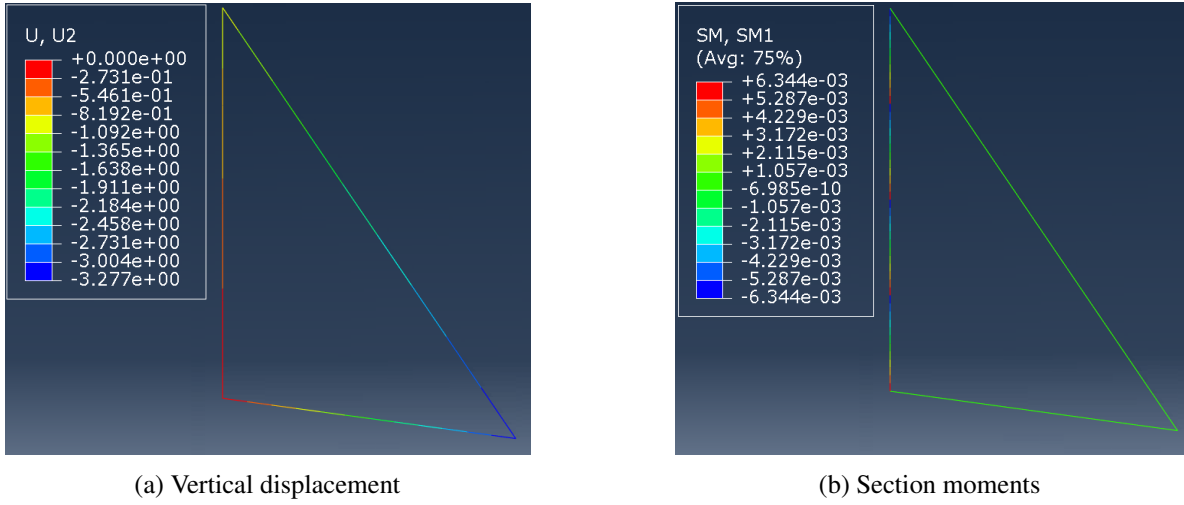


Figure 8: Triangular structure using connected beam elements

2.6 Result analysis: pin-pin beam

As can be clearly seen in (Fig. 5) and (Fig. 6) the number of elements can dramatically affect the results of a simulation, thus is important to choose wisely to find a balance between precision and simulation time. In (Fig. 5a) and (Fig. 6a) since the beam is simulated with only two elements, all the distributed load is transformed in an equivalent nodal load just in the point in the middle of the beam. In this way the structure will show a behaviour similar to a beam loaded with a concentrated load in the middle, as the deformed shapes in the pictures suggest. Increasing the element number will separate more evenly the distributed load into more equivalent nodal forces, showing a more correct representation of the problem, with a "rounder" deformed shape in (Fig. 5b) and (Fig. 6b).

From the theory is easy to compute the maximum deflection of the simply supported beam and it's clear that it perfectly matches the results coming from the FEM analysis in both cases, while the section moment is matched only in the case with more elements present.

$$\delta_{max} = \frac{pL^4}{384EI} = -0.0152mm \quad M_{max} = \frac{pL^2}{8} = -1250Nmm \quad (2.1)$$

2.7 Result analysis: triangular structure

From (Fig. 7) is clear to see that the results for the vertical displacement are the same, while the section moments have very high values, but they should not be present. Since the original structure was a closed structure, just changing the element from truss to beam will not give the same results. Using welded truss elements it's like putting hinges at each node, while beam elements will allow moments to be transferred between them, as can be seen in (Fig. 7b). To overcome this problem the structure was re-modelled as described in (Sec. 2.3), in this way the beam elements were internally connected by hinges, simulating the same structure of Ex.1, now the elements will not transfer moments and thus will act as trusses. The results can be seen in (Fig. 8), where the displacements are not changed, while the section moments are present, probably due to numerical calculations, but very close to zero, correctly representing the behaviour of a beam connected with hinges simulating a truss.

3 Ex 3: Machine tool with beam elements

3.1 Scope of the exercise

The scope of this exercise was to carry out a preliminary stiffness analysis of a complex machine tool structure, using beam elements, to verify if the stresses acting on the structure are smaller than the critical ones, in order to choose the correct material for the structure.

3.2 Model description

3.2.1 Geometry

To run a preliminary stiffness analysis of the machine tool the geometry was highly simplified: starting from the initial problem (Fig. 9a) the model was broken down to just three two dimensional beams with their own geometry and sections shown in (Fig. 9c), while the dimensions are shown in (Fig. 9b). The two vertical beams are representing the columns of the tool, which support the crosshead, deform axially, in bending and in torsion, and the horizontal beam represents the crosshead, where the tool head moves, which undergoes bending and torsion. The material used is standard steel (Tab. 4).

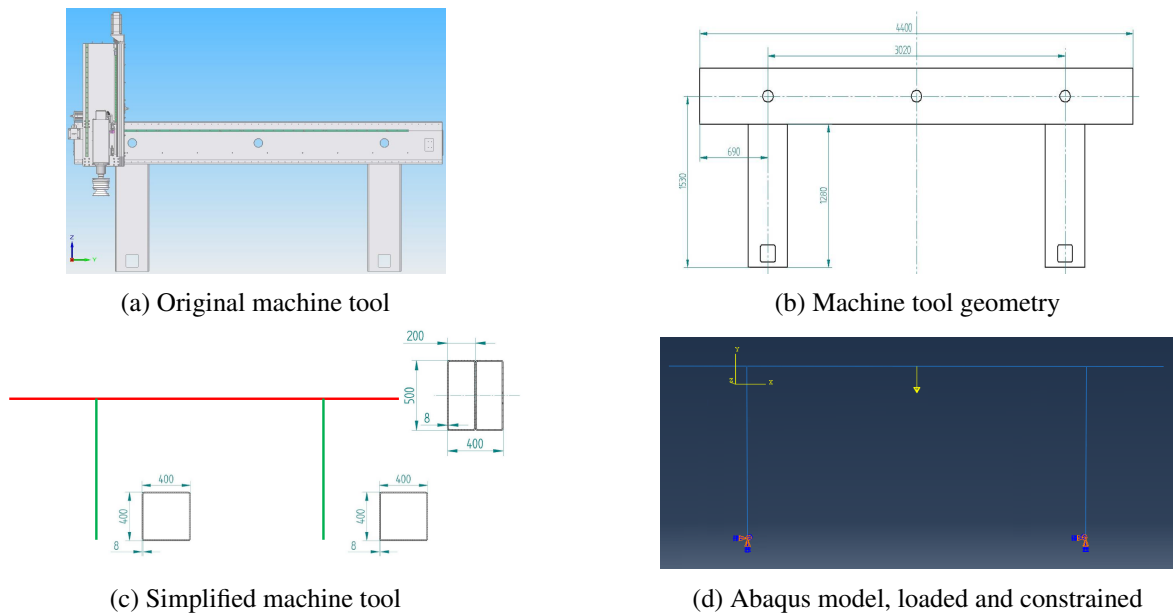


Figure 9: Model geometry

Due to the peculiar shape of the crosshead's section, it was not possible to insert the dimensions of the cross-section and let Abaqus compute the needed data, but a generalized profile was created and the values of area A , second order area moments J_{11} and J_{22} , and the torsion constant J_t were inserted manually into Abaqus. The area and second order moments were computed as the value of the whole section minus the empty part, while the torsion constant value was computed with (Eq. 3.1).

$$J_t = \frac{16b^2h^2t}{4b+h} \quad (3.1)$$

The data inserted in the generalized profile are in (Tab. 4) as well as the steel material data.

A	21888 mm^2	E	206000 MPa
J_{11}	$6.9 \cdot 10^8 \text{ mm}^4$	ν	0.33
J_{22}	$3.83 \cdot 10^8 \text{ mm}^4$		
J_t	$9.85 \cdot 10^8 \text{ mm}^4$		

Table 4: Section data

3.2.2 Boundary conditions

To properly constrain the structure, the two columns were connected to the ground using two en-casters, blocking rotations and displacements in every direction, as shown in (Fig. 9d). The columns and crosshead were effectively created as a unique part, practically welding them together.

3.2.3 Load

The load considered for this problem is the gravitational force of the tool head, which can be regarded as a concentrated force equal to 500 N acting on the crosshead. In order to consider the worst case scenario, the load is placed at the centre of the crosshead, since here is where the maximum deflection will take place, (Fig. 9d).

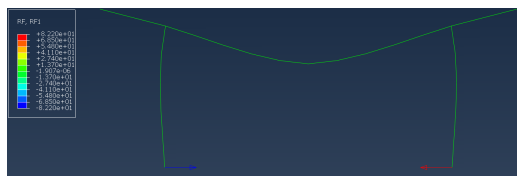
3.2.4 Mesh

The mesh was created with 24 elements, using the Abaqus command "approximate global size" and leaving it set with the value of 300. The element type chosen was the beam, at first using the cubic deformation one, then it was changed in the shear-flexible one (shear-flexible beam). The beam element was chosen since, in first approximation, both the crosshead and the columns will not deform axially.

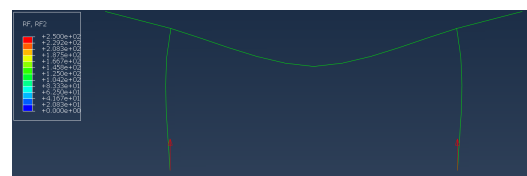
3.3 Simulation results

3.3.1 Reaction forces

Using both beam element types the resulting reaction forces were the same in each case, showing that the problem was well posed. As it could be expected the horizontal reaction forces (Fig. 10a) acting on the columns clamps, which result from the deflection happening at the connection between the columns and the crosshead, were equal and opposite, with a value of $RF_1 = 82.2 \text{ N}$. The vertical forces (Fig. 10b) instead were equal both equal to half of the load $RF_2 = 250 \text{ N}$ and in opposite direction, which is correct since the load was applied in the symmetry plane of the problem.



(a) Horizontal reaction force



(b) Vertical reaction force

Figure 10: Resulting reaction forces

3.3.2 Vertical displacement

Vertical displacement is probably the quantity where the difference between the beam's types is more clear. Both types give the expected result: maximum vertical displacement at the middle of the crosshead, where the load is applied. Nevertheless, the values in play are slightly different, it is clear that the shear-flexible beam (Fig. 13b) allows for greater displacements, resulting in a less stiff element.

δ_{max}^{cubic}	0,001205 mm
$\delta_{max}^{shear-flex}$	0,001561 mm

Table 5: Maximum beam deflection

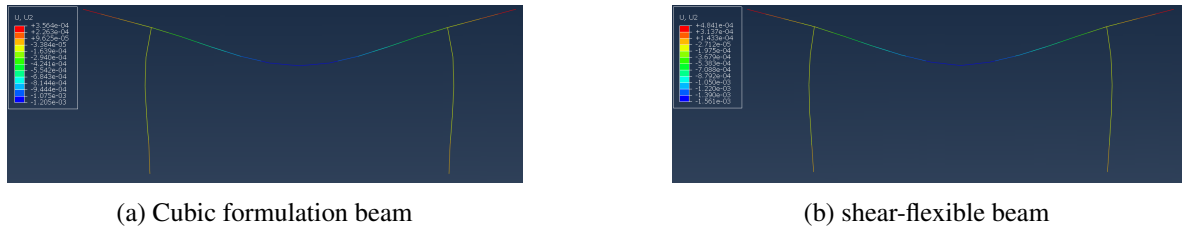


Figure 11: Resulting vertical displacement

3.3.3 Stresses

Looking at stresses a clear distinction can be made once again between cubic formulation beam and shear-flexible beam, with the latter one being less stiff, resulting in smaller stresses. Another important observation can be done by comparing (Fig. 12) and (Fig. 13), the only difference between these pictures is where the stresses are computed. The beam element used have a section of non negligible dimension so, depending on where the computations are carried out, the results will be different as the beam will deform more on the side opposed to the one where the load is applied, stretching more and resulting in higher stresses, highlighting the importance of selecting correctly the position where to compute stresses. In this case the beam will deform more on the bottom side, leading to higher stresses.

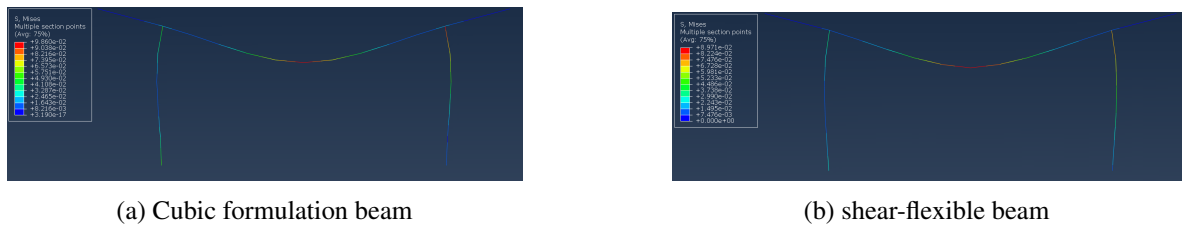


Figure 12: Stresses evaluated at bottom

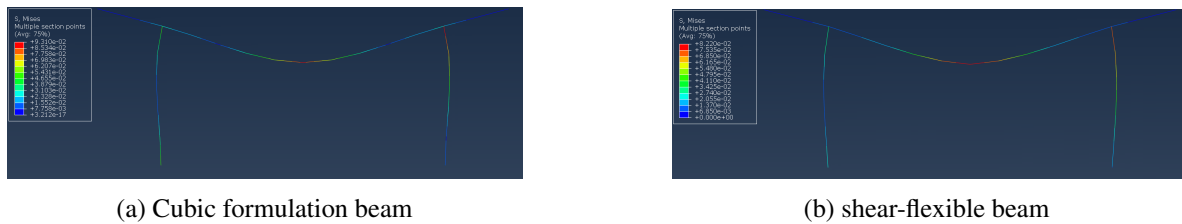
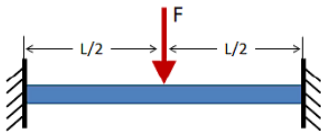


Figure 13: Stresses evaluated at top

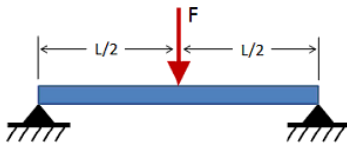
3.4 Result analysis

3.4.1 Analytical approach

To verify the validity of the results obtained with Abaqus is necessary to compare them with analytical results. To achieve that, the geometry of the problem is further simplified by transforming the two columns into equivalent constrains, bringing the problem to a simple single beam with two constrains. The columns offer some resistance to torsion and bending, but at the same time they also bend and compress themselves, offering some room for displacement; thus their behaviour will be in the middle between acting as two clamps and acting as two pins. Following this reasoning the maximum deflection coming from Abaqus should lay between the maximum deflection given by the double clamp case and the double pin one. The values are reported in (Eq. 3.3) and (Eq. 3.1). The actual value coming from Abaqus is $0,001561\text{ mm}$ or $0,001205\text{ mm}$ (depending on element type), which is clearly closer to the double pin case, which actually make sense because, by looking at the deformed shape of the entire structure, is clear that the columns allow deflection and offer small resistance, something that a clamp cannot do. So, even if it is not possible to define if the results are exact, at least they are bound between approximate situations, giving a high degree of confidence that the results are correct.



$$\delta_{max} = \frac{FL^3}{192EI} = 0,0005\text{ mm} \quad (3.2)$$



$$\delta_{max} = \frac{FL^3}{48EI} = 0,002\text{ mm} \quad (3.3)$$

3.4.2 Cubic formulation versus shear-flexible

The difference between the two types of beam can be easily analyzed in this problem. The cubic formulation uses the Eulero-Bernulli approximation that the cross sections of the beam remain perpendicular to the neutral axis and in this way the deformations due to shear are neglected, resulting in a stiffer structure, this formulation is particularly effective for slender beams. In the shear-flexible beam this hypothesis is no longer used and the cross sections can rotate respect to the neutral axis, in this way also shear deformations are considered, thus the structure will deform more and will result less stiff than the Eulero-Bernoulli one. Shear-flexible beams are usually used for short and thick beams, but they are good also for slender beams, overall they are more general and can be used effectively for most of the problems.

In this problem being the beam section very big but with small thickness probably is best to use the shear-flexible approach, being more general and precise for non-slender beam. As shown in (Tab. 5) the shear-flexible shows a bigger vertical displacement, as expected from the theory.

4 Ex 4: Dynamic analysis

4.1 Scope of the exercise

Objective of this exercise was to analyze the static and dynamic response of a "L-shaped" beam structure, solving for the resonance frequencies and the mode shapes and comparing static and dynamic deformation. Solutions were found both analytically, given the structure simplicity, and numerically to compare them.

4.2 Model description

4.2.1 Geometry

The structure geometry is shown in (Fig. 14), the section and material elastic properties inserted into Abaqus are summarized in (Tab. 6).

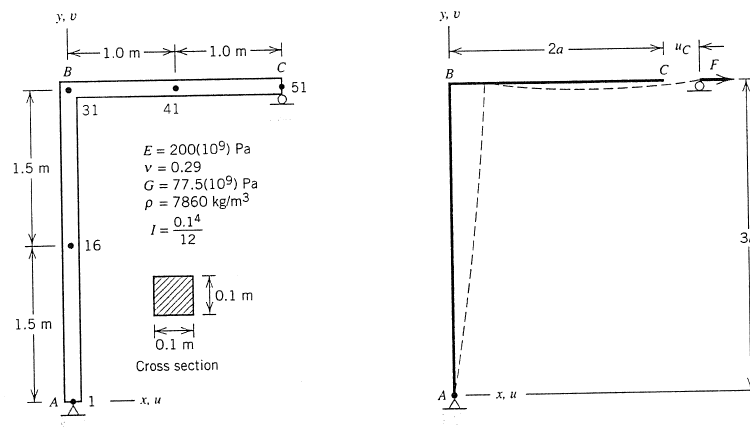


Figure 14: Structure geometry and load

E	200000 MPa	I	$8.333 \cdot 10^6 \text{ mm}^4$
ν	0.29	AB	3000 mm
G	77500 MPa	BC	2000 mm
ρ	$7.86 \cdot 10^{-9} \text{ ton/mm}^3$	h_{sec}	100 mm

Table 6: Structure data

4.2.2 Boundary conditions

To constrain the structure a simple hinge-roller setup was implemented. In this way the system is isostatic and can be easily analysed even analytically. The hinge was applied to point A and the roller to point C. Implemented in Abaqus in 2D by constraining horizontal and vertical displacement in A and vertical displacement in C (Fig. 15).

4.2.3 Load

The load was applied in node C as a positive horizontal concentrated force. Two cases were analysed:

- Static load: a constant 3000 N load was applied to study the structure's behaviour and deformation under a static load.
- Dynamic load: a time varying sinusoidal load was applied equal to $3000\sin(\Omega t)$ $[\text{N}]$ to study the structure's dynamic response to an oscillatory load, to also study its frequency response function.

4.2.4 Mesh

Element wise, it was chosen the B23 beam, which is a planar 2-node beam with cubic formulation. The chose element size was 100 mm , in this way problems with the single elements resonances was avoided and the nodes were coherent with the analytical formulation. In this way a 50 elements mesh was created.

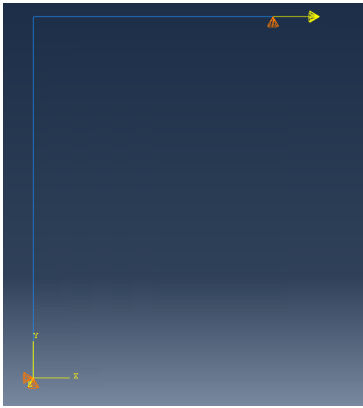


Figure 15: Load and boundary conditions

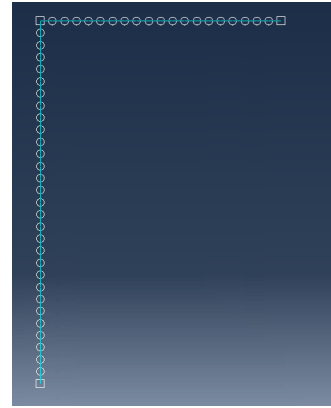


Figure 16: Mesh

4.3 Simulation results

4.3.1 Static analysis

After applying the static load of 3000 N at node C it was possible to study the structure response to this input, plotting the horizontal displacement in (Fig. 17). As one could expect, the bigger displacements appear near the loaded node, where the roller is present, while getting closer to the hinge the displacements tend to zero.

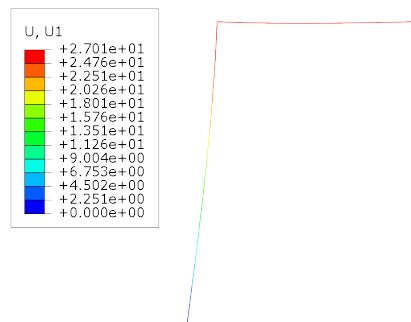


Figure 17: U_1 static load

4.3.2 Dynamic analysis

Applying the dynamic load it was possible to find the response of the structure at different frequencies. In particular, the modes shapes where computed finding the eigenvectors and eigenvalues of the dynamic problem plotting them in (Fig. 18). Since the structure is relatively simple it was possible to compare the analytical results with the numerical ones in (Tab. 7), where is clear that the higher the mode the higher the error, even if it always lays in reasonable values.

	Analytical	Numerical
Mode 1	3.315 Hz	3.307 Hz
Mode 2	35.08 Hz	35.075 Hz
Mode 3	70.77 Hz	70.971 Hz
Mode 4	122.7 Hz	123.540 Hz
Mode 5	226.0 Hz	228.430 Hz

Table 7: Resonance frequencies

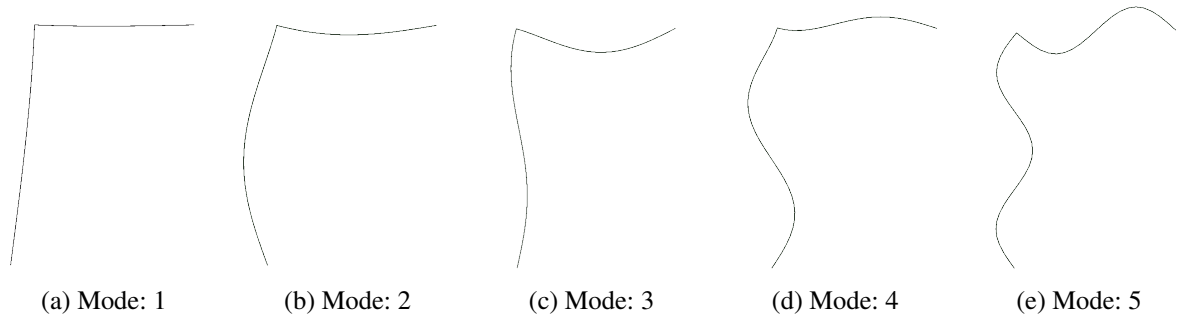


Figure 18: Mode shapes

4.4 Requests

4.4.1 Estimation of f_1

In order to estimate analytically f_1 the structure was simplified and transformed in an equivalent one where to carry on all needed calculations. The structure was divided in 2 elements, a vertical one and an horizontal one. Subsequently the masses of these 2 elements were lumped at their nodes assuming uniform mass distribution along the structure. Then assuming to apply a unitary force to node C it was possible to compute the structure deflection δ , knowing the lengths of the elements, by assuming an equivalent stiffness of the structure k_{eq} depending on the material and on the sectional moment of inertia. This in condensed into (Eq. 4.1), where m_1 and m_2 are the lumped masses of the elements and a is half the length of the horizontal element. The result is quite close the the analytical and numerical one, showing that it is a good estimation.

$$f_1 = \frac{1}{2\pi} \sqrt{\frac{k_{eq}}{m_2 + m_1/2}} = 3.1986 \text{ Hz} \quad \text{where} \quad k_{eq} = \frac{1}{\delta} \quad \text{and} \quad \delta = \frac{15a^3}{EJ} \quad (4.1)$$

4.4.2 Static versus dynamic comparison of u_1 at node 18

As requested in the exercise, the horizontal displacement u_1 of node 18 (the midpoint of the vertical beam) was sampled and compared between the structure under static load and under the dynamic load at $122Hz$, the values are reported in (Tab. 8). This great difference between the two values is given by the fact that node 18 results very close to a node of the deformed shape of the structure at $122Hz$, meaning that, while the structure is vibrating at $122Hz$, that point will move very slightly.

	u_1^{18}
Static	$16.5426mm$
Dynamic	$0.0285mm$

Table 8: Node 18 horizontal displacement

4.4.3 u_1 estimation of node 18 at $122Hz$

Due to the simplicity of the problem it was possible to estimate by hand calculations the horizontal displacement u_1 of node 18 at $122Hz$, studying the structure at its fourth mode shape $f_4 = 123.54Hz$. The structure, with its constraints, is effectively a one degree of freedom system, able of only moving to the right. To compute u_1 the nodal equations $\{D\} = [\phi]^T \{z\}$ were used, knowing the applied load p_i is possible to the modal coordinates z and using the modal matrix $[\phi]$, the final displacements $\{D\}$ containing u_1 could be found.

$$p_4 = u_1^{(3)} \cdot P = 0.684531 \cdot 3000N = 2053.6N \quad (4.2)$$

$$\omega_4 = \frac{2\pi}{f_4} = \frac{2\pi}{123.54Hz} = 776.2rad/s \quad (4.3)$$

$$z_4 = \frac{p_4}{\omega_4^2} \cdot 2\xi = \frac{2053.6}{776.2^2} \cdot 20.02 = 0.0852 \quad (4.4)$$

$$u_1^{(18)} = z_4 \cdot U_1^{(18)} = 0.0852 \cdot 0.880017 = 0.075mm \quad (4.5)$$

The load was computed in (Eq. 4.2) where $u_1^{(3)}$ is the displacement of the node 3, where the load is applied, at the fourth vibration mode, computed in Abaqus. The result coming from Abaqus, computed as the module of the complex number of $u_1^{(18)}$, is equal to $0.0634mm$. The slight difference respect to the value computed in (Eq. 4.5) is due to the fact that the calculations are made at slightly different frequencies.

4.4.4 Static versus dynamic comparison of RF_1 at node 1

The comparison between static and dynamic load was made sampling the reaction forces at $122Hz$. The reaction force in the static case is equal to the applied load, since it is the result of an external force balance. Instead, the result of the dynamic load is the sum of the external load with also the internal force deriving from the deformation of the structure.

	RF_1
Static	$3000N$
Dynamic	$2065.33N$

Table 9: Node 1 horizontal reaction force

5 Ex 5: Plate with a hole

5.1 Scope of the exercise

The main objectives of this exercise were to carry out a convergence analysis and study how to build meshes using different techniques on a holed steel plate.

5.2 Model description

5.2.1 Geometry

The studied geometry was a square steel plate of dimensions $[10 \times 10 \times 1] \text{ mm}$ with a 7 mm radius hole in the centre, as shown in (Fig. 19a). In order to simplify the problem and shorten the computation time the domain was split along its symmetry planes and only a quarter of the plate was analysed (Fig. 19b), maintaining the correct dimension and applying a symmetry boundary conditions to the split end. The material used has its mechanical elastic characteristics shown in (Tab. 10).

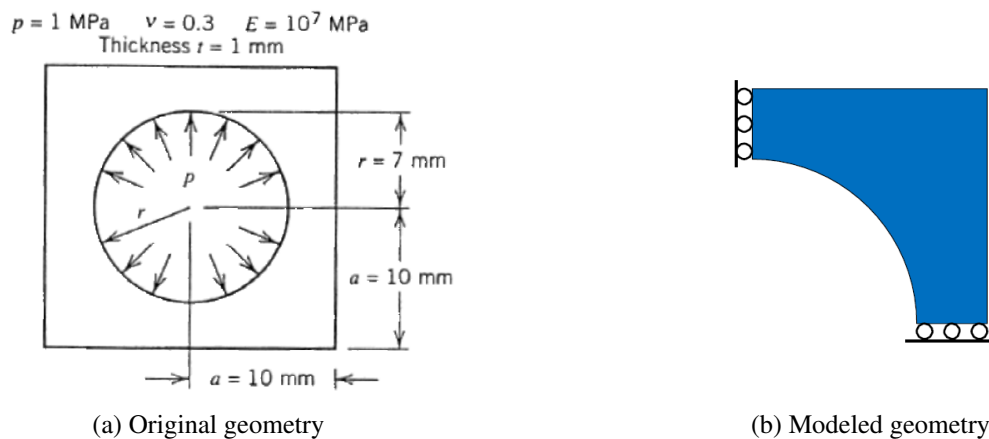


Figure 19: Problem geometry

a	10 mm	p	1 MPa
r	7 mm	E	10^7 MPa
t	1 mm	ν	0.3

Table 10: Model data

5.2.2 Boundary conditions

To correctly implement the problem geometry, two symmetry boundary conditions were imposed at the edges where the domain was split, along the x direction and the y direction, namely in the top-left part and the bottom-right, as shown in (Fig. 19b). It was chosen not to use the symmetry boundary condition Abaqus provides but instead the single degrees of freedom were constrained. In particular, the constraints were applied in the top-left edge to the horizontal and rotational dofs u_1 and ur_3 , while in the bottom-right edge to the vertical and rotational dofs u_2 and ur_3 .

5.2.3 Load

The load imposed from the problem is a distributed pressure load applied to the internal part of the hole. It was implemented in Abaqus by applying a pressure load p to the curved edge of the hole, with the value in (Tab. 10).

5.2.4 Mesh

Many different meshes were created, at first a rather coarse and not well structured mesh was used to analyse the behaviour of different element types with this type of mesh, then 3 different meshes were created with the same element but with different refinement to carry out a convergence analysis.

5.2.4.1 Element type analysis mesh

After trying some different options to create the mesh, it was chosen to run the simulations with the mesh in (Fig. 20). This mesh had a 1 mm seed, a structured mesh algorithm for a five logical edge structure was used, with only quadratic elements. The mesh is not perfect as some elements are quite distorted, in particular along the loaded edge, this will create some problems especially with simpler elements. Three simulations were run using this mesh but with different element types:

1. 4-node bilinear plane stress quadrilateral element, with reduced integration and hourglass control.
2. 4-node bilinear plane stress quadrilateral element, with full integration.
3. 8-node biquadratic plane stress quadrilateral element, with full integration.

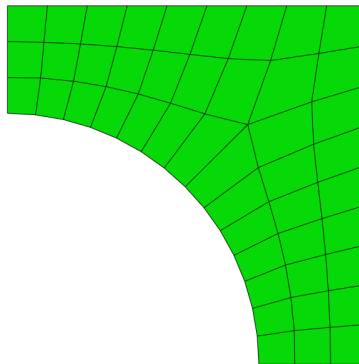


Figure 20: Coarse mesh

5.2.4.2 Convergence analysis mesh

Then, to run a convergence analysis, a new mesh was created. The structure was partitioned, effectively dividing it in half by putting a node at 45° along the quarter circumference. In this way the regions to mesh were smaller and it was easier to mesh the part using a structured mesh, since the regions to mesh had only four logical edges. The simulation was run with different mesh size (Fig. 21):

1. 4-node bilinear plane stress quadrilateral element, full integration, 1 mm seed
2. 4-node bilinear plane stress quadrilateral element, full integration, 0.5 mm seed
3. 4-node bilinear plane stress quadrilateral element, full integration, 0.25 mm seed

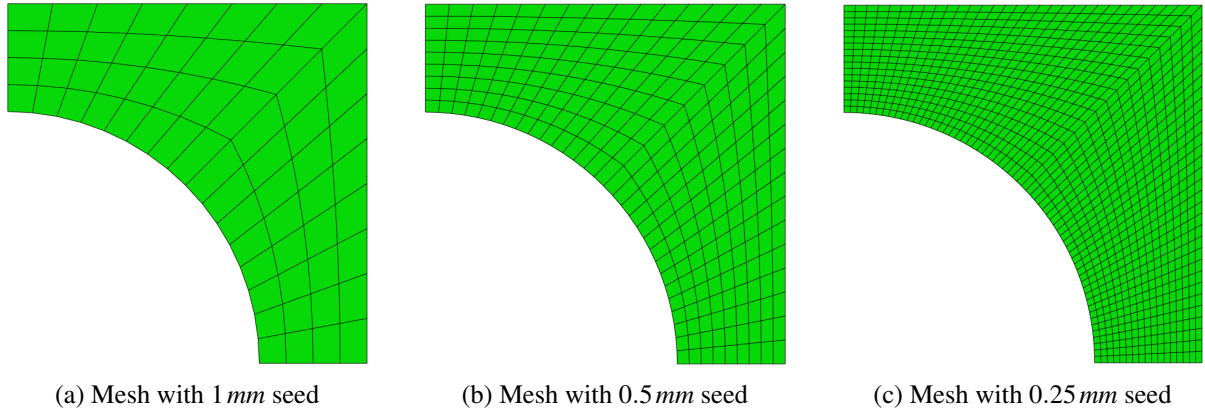


Figure 21: Convergence analysis mesh

5.3 Simulation results

Analysing Von Mises stress it was possible to highlight the difference between the different mesh element types (Fig. 22) and for different mesh refinement (Fig. 23). In both cases the results are not averaged along the elements, to get a better grasp of the simulation behaviour.

5.3.1 Element type analysis mesh

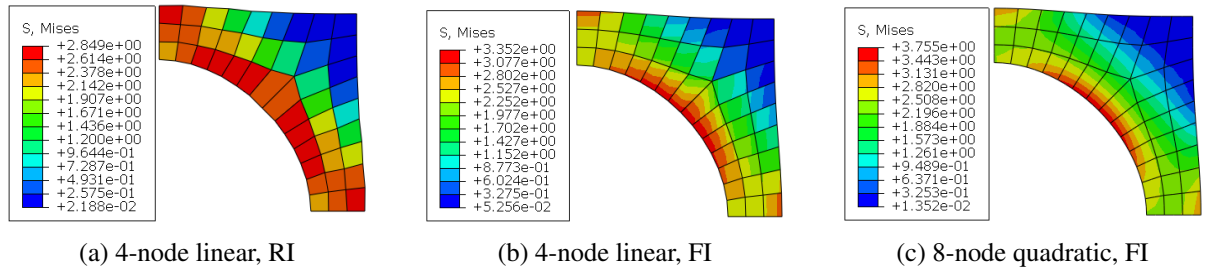


Figure 22: Von Mises stress

5.3.2 Convergence analysis mesh

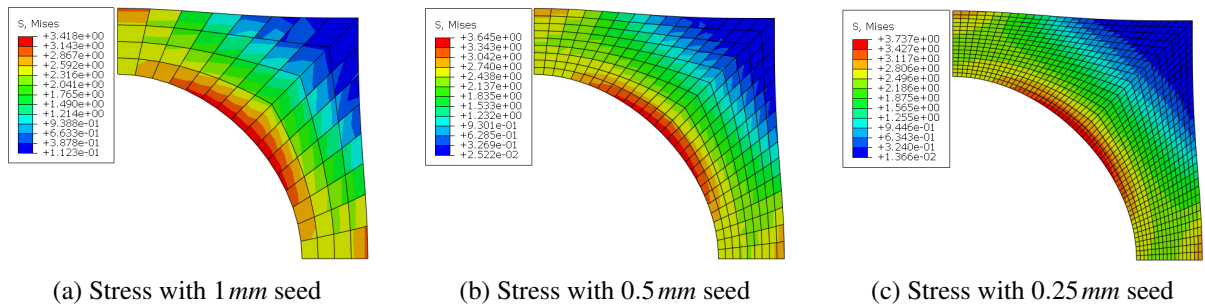


Figure 23: Von Mises stress convergence

5.4 Result analysis

5.4.1 Element type analysis

Analyzing the plots in (Fig. 22) it is possible to grasp the different behaviour, precision and validity of the different element types.

1. The 4-node bilinear plane stress quadrilateral element, with reduced integration (Fig. 22a) clearly showed its limitations with this coarse mesh. The two elements placed at 45° along the circular edge were way to big and distorted, this would be acceptable if they were far from the load, but instead one of their edge is directly loaded by the pressure, this created some errors. In particular, being loaded on one edge, the nodes of this edge resulted in having very high stresses, while the other nodes, which were very far from the loaded edge, due to the elements dimensions, resulted in having way lower stress. Using reduced integration, the resulting element stress is an average of the nodal stresses, resulting in a lower than expected result. Indeed it makes no sense to have the central elements of the circumference to have lower stresses than their neighbouring elements, as they should be the more loaded ones. Thus this discretization, with this element type, is wrong.
2. 4-node bilinear plane stress quadrilateral element, with full integration (Fig. 22b) were an improvement respect to the previous element. Using full integration with four integration points it was possible to some variation of stresses in the single elements, but still the simulation was not precise. Again in the two elements at 45°, which should be more loaded, the regions with higher stresses were smaller compared with their neighbouring elements, showing that the simulation was still not correct since there it was expected to have the highest stresses.
3. The 8-node biquadratic plane stress quadrilateral element (Fig. 22c), with full integration was clearly the better element type, thanks to its eight nodes it was able to highlight better the stresses distribution in the structure, with smoother transitions between elements, even with a very coarse mesh. As it was expected the highest stresses could be found in the middle of the circumference, while the lowest ones are found far away in the corner, where the most material is present. This precision came to a cost of higher computational time.

5.4.2 Convergence analysis

In (Fig. 23) results for the same element but with different mesh size are presented and the stress distributions analysed. By varying the mesh it was possible to notice that the finer the mesh the more precise the results and the smoother the transitions between the stress levels.

Since the mesh nodes were preserved, the evaluation point for the stress was always the same and the element type was not varied, a convergence analysis could be carried out. To study mesh convergence the order of the error was estimated, by decreasing the mesh element size the stress got higher, and the trend is expected to be linear. Using the dimensionless parameter h (Eq. 5.1), plotted against the stress (evaluated always at the same point but for different meshes), the trend is expected to be linear, with smaller error for finer mesh.

The error can be computed as follows (Eq. 5.1), where σ_∞ is computed from the interpolation line or can be computed as (Eq. 5.2).

$$err = \frac{\sigma_\infty - \sigma_i}{\sigma_\infty} \cdot 100\% \quad h = \frac{1}{\sqrt{N_{elem}}} \quad (5.1)$$

$$\sigma_\infty = \frac{\sigma_1 h_2^1 - \sigma_2 h_1^1}{h_2^1 - h_1^1} \quad (5.2)$$

As can be seen in (Fig. 24), with three different mesh refinement it was possible to interpolate the data and extract a function which describes the data behaviour. As expected the trend was linear, since $R^2 = 0.9982$, and $\sigma_\infty = 3.8545 \text{ MPa}$ can be found as the constant part of the linear trend. This linear trend shows that the mesh setup was carried out successfully, and that, in the case of an infinitesimally small element, the stress computed would be the most precise stress. The fact that the discretisation error is linear does not mean that the result one can obtain is the correct one, as errors in the load application on in the boundary conditions could be made. In this case, the discretisation error can still be linear, but the results wrong.

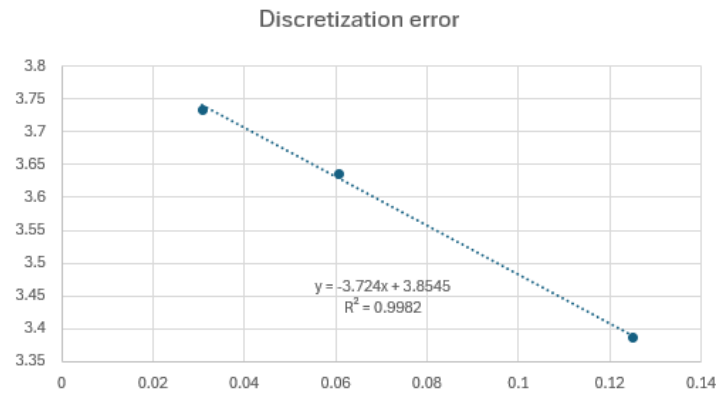


Figure 24: Discretization error

6 Ex 6: Plasticity

6.1 Scope of the exercise

Scope of the exercise was to investigate the effects of plasticity on a simple geometry, similar to the one of (Ex.5). Both the loading and unloading phase were studied, to investigate how the geometry changed after the cycle and to find when the yielding stress was surpassed, thus plasticity happened.

6.2 Model description

6.2.1 Geometry

The analysed geometry was a square plate of $[100 \times 100] \text{ mm}$ with a 20 mm radius hole in the centre. Thanks to symmetry only one quarter of the plate was actually modelled in Abaqus (Fig. 25).

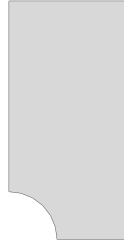


Figure 25: Studied geometry

The material used was Al 2024 T3 considering both its elastic and plastic behaviour. The Young's modulus was set to $E = 73800 \text{ MPa}$, while the Poisson's ratio was set to 0.3333 being a metal. Laboratory data were used to characterize the plastic behaviour of the material, the data put into Abaqus are the true stresses over the true plastic strain of the material, computed as (Eq. 6.1) and collected in (Fig. 27), where the first stress is actually the yielding stress σ_Y , after which the material starts to yield and deforms plastically.

$$\varepsilon_{tr,pl} = \ln(1 + \varepsilon) - \frac{\sigma(1 + \varepsilon)}{E} \quad (6.1)$$

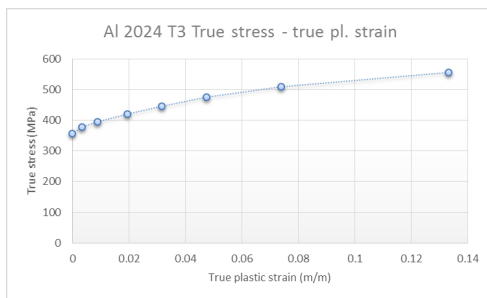


Figure 26: True stress over true plastic strain

True stress [MPa]	True plastic strain [1]
355	0
356.865525	0.000405664
378.208125	0.003393842
395.616	0.008936654
420.414	0.019386114
446.555	0.031726484
474.93	0.047484476
509.527	0.07384599
557.1808	0.133255001

Figure 27: True plastic strain

6.2.2 Boundary conditions

To correctly simulate the structure, and constrain all the rigid motions, two symmetry boundary conditions were applied to the two edges where the structure was split. This resulted in the top-left edge being constrained along u_1 and ur_3 , while in the bottom-right edge along u_2 and ur_3 .

6.2.3 Load

A uniform pressure load of -250MPa was applied to the upper edge of the plate, the minus sign is due to Abaqus conventions of load applied to edges. Two steps were created, one for the loading phase and one for the unloading one, in both the `Nlgeom` command was not used, since no large displacements were expected and the non linearities coming from plasticity were already accounted for during the material definition. During the unloading phase, the load applied was equal in modulus but with opposite versus respect to the loading phase, in this way the loads will cancel out and the structure will be unloaded when the step finished.

The increment size of both steps was reduced to 0.1, in this way the loading was not immediate but happened in more steps, in this way it was possible to easily identify when plasticity started to occur.

6.2.4 Mesh

The mesh used in the following simulations (Fig. 29) was created with a 3mm seed, using a quadratic structured mesh with $2D$ quadratic, full integration elements. This resulted in the creation of 488 elements and of a acceptable mesh, with the elements positioned around the hole that have good characteristics and shape.

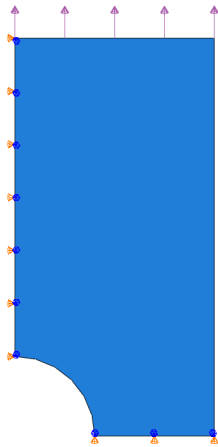


Figure 28: Loaded geometry

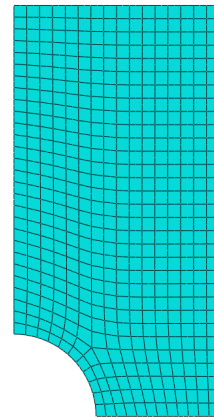


Figure 29: Meshed geometry

6.3 Simulation results

Interesting results to analyse are the Von Mises stress, to see how the plasticity affects the structure after unloading and the equivalent plastic strain (PEEQ) to study the regions permanently deformed by the plastic behaviour of the material.

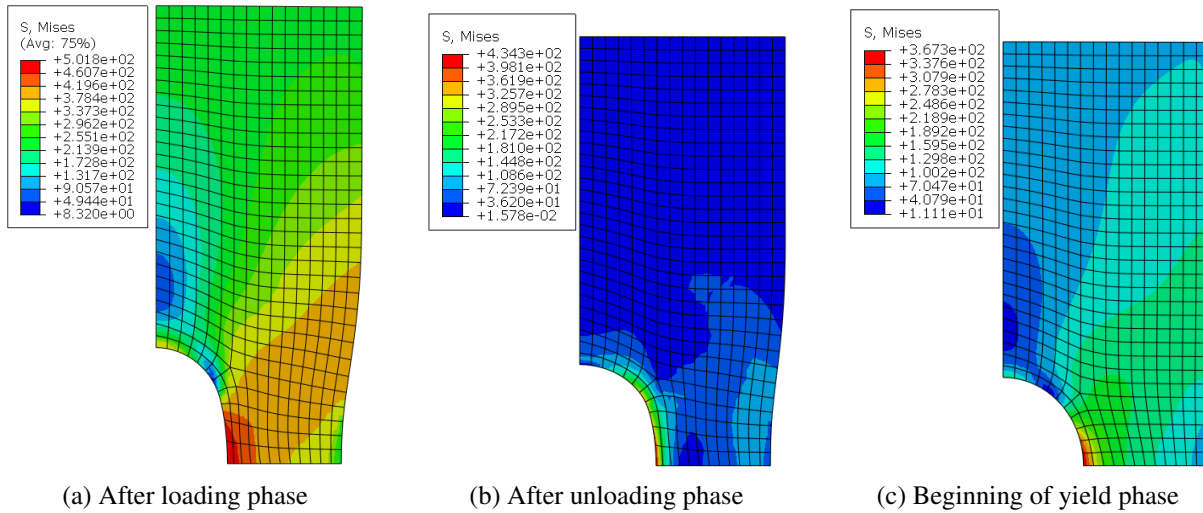


Figure 30: Von Mises stress

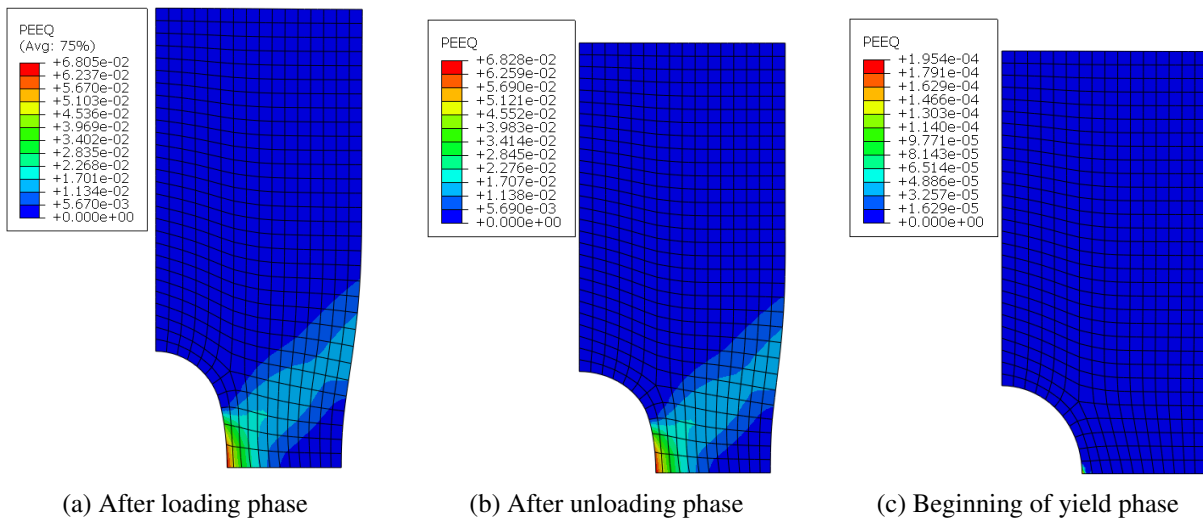


Figure 31: PEEQ

6.4 Result analysis

From the figures shown in (Sec. 6.3) it is possible to extract various informations, such as resulting stress and the start of the plasticity phenomena.

Studying (Fig. 30) it is clear to see that, after the loading phase where the yielding limit was surpassed, due to plasticity phenomena, the structure will not return to its undeformed shape but it will be subject to some degree of plasticization and deformation, leading to the presence of residual stresses (Fig. 30b). In (Fig. 30c) represents the starting point of plasticization phenomena since the maximum stress experienced at the bottom of the hole is equal to 367.3 MPa which is greater than the yielding stress $\sigma_Y = 355 \text{ MPa}$. This happens when the load applied is almost 40% of its maximum.

Looking at (Fig. 31) instead is possible to individuate where the plastic deformation occurs. Between the loading and unloading phase the values do not change, since the stresses inside the material are lower than σ_Y , but of course the deformed shape will change due to the elastic behaviour of the material. In (Fig. 31b) it is possible to find the region more plastically deformed, which is the centre part of the plate, where less material is present to oppose the deformation, thus will deform more than the rest and will be subject to higher stresses, even higher then the yielding stress. This region will be the one more influenced by residual stresses. In the bottom left of (Fig. 31c) is present the first region of the plate where plasticity starts, which is indeed the same region of (Fig. 30c) where the stresses surpass σ_Y .

The overall plate mirrored thanks to the symmetry boundary condition.

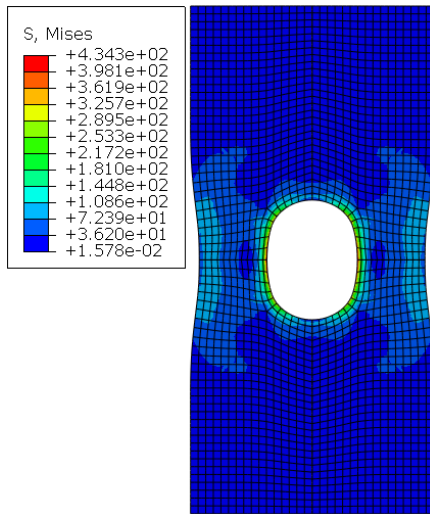


Figure 32: Von Mises stress

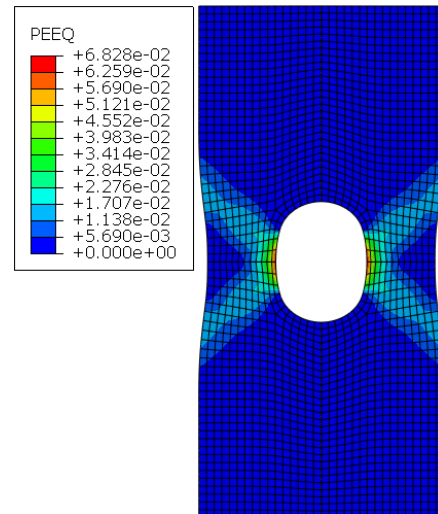


Figure 33: PEEQ

7 Ex 7: Lateral-Torsional instability

7.1 Scope of the exercise

Scope of the exercise was to investigate the behaviour of a beam undergoing lateral-torsional instability. The beam had a high torsional stiffness in the direction of the load, but due to the high compressive stresses it underwent buckling, resulting in a twist of the cross-section, in a direction it was not designed for, resulting in unwanted large displacements in lateral direction.

7.2 Model description

7.2.1 Geometry

The studied geometry consisted in a beam long $L = 2000\text{ mm}$, with a cross-section presented in (Fig. 34). The geometry was created drawing the centreline of the cross-section, which was then extruded for the required length and the section used was a continuous homogeneous shell with thickness 5 mm . The material used was a steel with mechanical elastic behaviour, using $E = 210\text{ GPa}$ and $\nu = 0.33$.

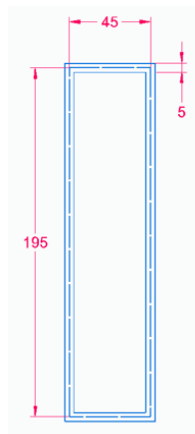


Figure 34: Beam's cross-section

7.2.2 Boundary conditions

The beam was encastered at one of the ends, constraining each degree of freedom. The boundary condition was applied to the whole beam's end surface, visible in the back of (Fig. 35).

7.2.3 Load

A load $P = -1\text{ N}$ was imposed at the free end's top side middle point. Therefore it was needed to create a reference point RP in that position, to actually apply it, as shown in (Fig. 35). Then a coupling between the reference point and the free end's surface was created in order to correctly apply the load.

The load was put equal to a unitary value since buckling analysis will give as output a multiplier that measures how far the applied load is from the critical load inducing buckling. Therefore by using a unitary load the simulation will output directly the critical value for which instability will occur.

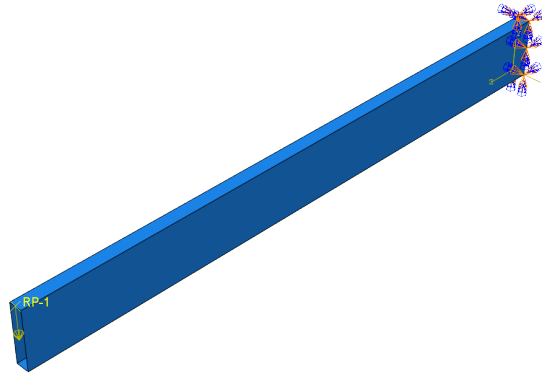


Figure 35: Beam's load and boundary conditions

7.2.4 Mesh

The element used was a quadratic element with 8 nodes and reduced integration (S8R), with a seed size of 25, resulting in the creation of a 1600 elements structured mesh (Fig. 36).

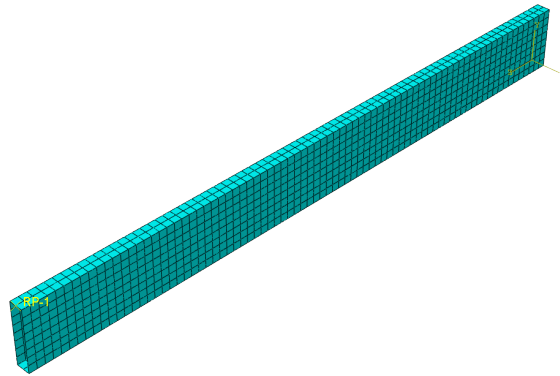


Figure 36: Beam mesh

7.2.5 Step

The step used was linear perturbation buckling type, with a maximum number of iterations to find the eigenvalue equal to 80 in order to achieve a solution.

7.3 Simulation results

The critical load computed for the first buckling mode resulted equal to:

$$P_{crit} = 189.2 \text{ kN} \quad (7.1)$$

In the figures below the first buckling mode for this problem is presented. It's clear to see that instability occurs by analysing (Fig. 37b) and (Fig. 37c), since the beam appears to deform more in lateral direction than in the vertical one, even though the original load was applied along the vertical direction.

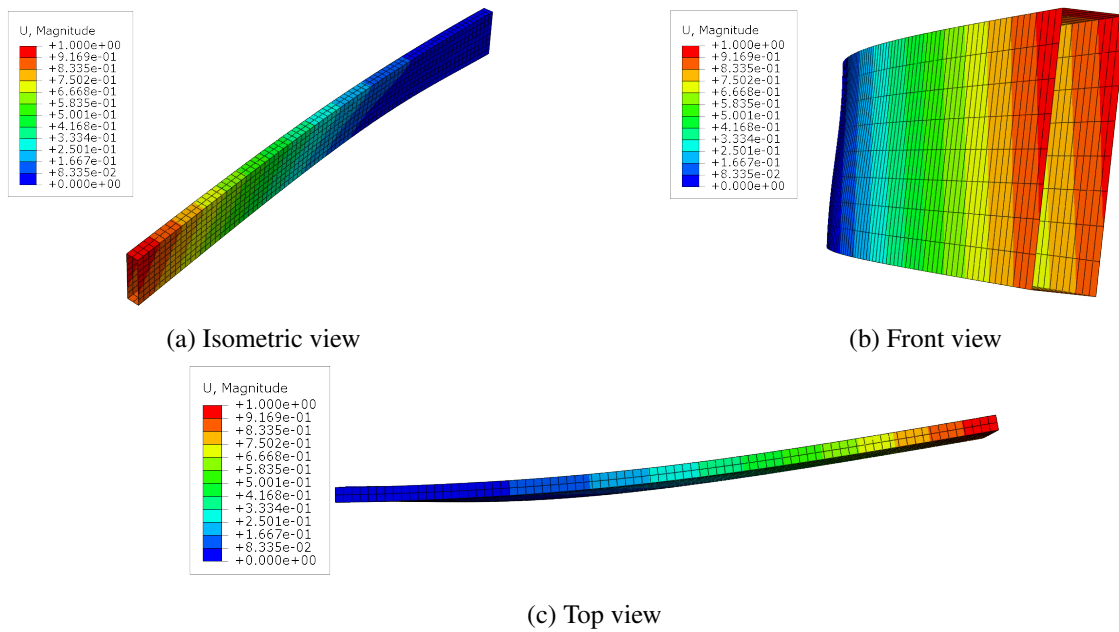


Figure 37: First buckling mode

7.4 Result analysis

7.4.1 Analytical result

To check if the simulation setup is correct it was possible to compare the results with an analytical solution for a similar problem shown in (Eq. 7.2). Where the critical load could be computed, after finding the section's moment of inertia presented in (Tab. 11).



$$P_{crit} = \frac{4.2}{L^2} \sqrt{E J_y G J_t} = 253.35 \text{ kN} \quad (7.2)$$

E	210000MPa	J_y	1070000mm ⁴
G	78752MPa	J_t	3290000mm ⁴

Table 11: Section data

As can be seen from (Eq. 7.1) and (Eq. 7.2), the values between the simulations and the analytical solution are in the same order of magnitude but are quite different.

This can be explained by the different point where the load is imposed: in the analytical solution the load is imposed at the centre of mass of the section, while in the simulation the load was imposed at the top edge. This means that when the beam deforms and twists, the load in the simulation will have an eccentricity respect to the centre, which will result in an additional resulting torque that will further increase the twisting of the section. Therefore, a smaller load P will be required to reach instability, explaining why P_{crit} of the simulation is lower than the analytical one.

7.4.2 Beam element

To verify the reasoning carried out in (Sec. 7.4.1), a second simulation was run, using beam elements instead of 2D ones. Thus, a 3D wire with a section resembling (Fig. 34) was created: the mesh consisted of 40 elements with cubic beam formulation.

All properties such as material, boundary conditions, step, etc... were the same as for the previous beam. The only difference was the position of the load, which now can only be applied to the end node, therefore resembling more the analytical case.

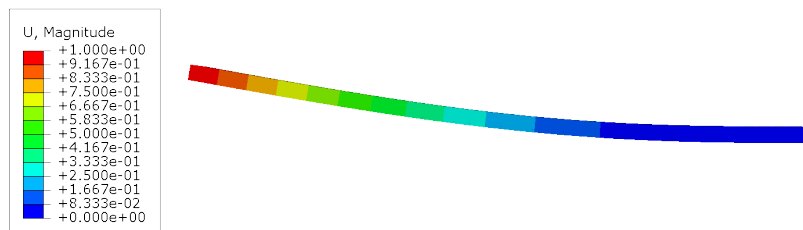


Figure 38: Beam element displacement

Again in this situation, even if harder to see, the beam deforms more in lateral direction than in the vertical one, showing again that lateral-torsional buckling instability occurs, while twist is much more contained, but still present.

The resulting critical buckling load is equal to:

$$P_{crit} = 238.3 \text{ kN} \quad (7.3)$$

This value is way closer to the analytical one (Eq. 7.2) since the load is applied in the center of the beam, as is in the analytical case.

7.4.3 Conclusion

In conclusion, lateral-torsional buckling analysis is pivotal in safety studies. Usually high safety factors are considered, at $\eta \approx 4$, to be sure that a structure can withstand the applied loads even in direction different from the one of the load, which may be not designed for. Since torsional buckling can happen even at regularly sustainable loads, a structure should be designed to avoid it. Therefore, a beam structure should have a section capable of resisting loads in the direction where loads are expected to happen, but also have a reasonably high moment of inertia in the directions where loads are not expected to happen. This is why, in general, perfectly rectangular thin beams are avoided, with a section as (Fig. 34), but always have some kind of flanges to have some material countering deflections in other unwanted directions.

8 Ex 8: Plate elements

8.1 Scope of the exercise

Scope of the exercise was to study and analyse a 3D steel plate coming from a .STEP file. The original file was analysed using 2D shell elements with thickness equal to the 3D plate's thickness, instead of 3D elements.

8.2 Model description

8.2.1 Geometry

The original 3D geometry is shown in (Fig. 39), the squared plate's dimensions are reported in (Tab. 12). To use 2D elements to describe the plate, instead of the 3D ones, the mid-plane surface of the plate was extracted in Abaqus to be used as surface upon which the 2D shell elements could be created.

The section used was a shell homogenous one, with section equal to the plate's one, meaning $t = 10\text{ mm}$. The step used was a static, linear perturbation one.

l	2000 mm	P	0.0005 N/mm ²
h	2000 mm	E	206000 MPa
t	10 mm	ν	0.333

Table 12: Section and material data

8.2.2 Boundary conditions

The plate was constrained via a simple support on the two vertical edges of (Fig. 39). To simply support the plate means to prevent it to move in the vertical direction, so constraining u_3 . In order to block all rigid translations, a second cornering node was completely constrained, locking also u_1 and u_2 . Then, to prevent also the remaining rigid rotation around z , another corner was locked along u_1 , as can be seen in (Fig. 40).

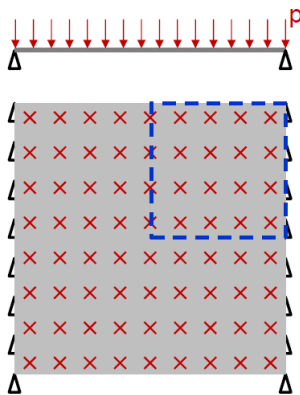


Figure 39: Original geometry and load

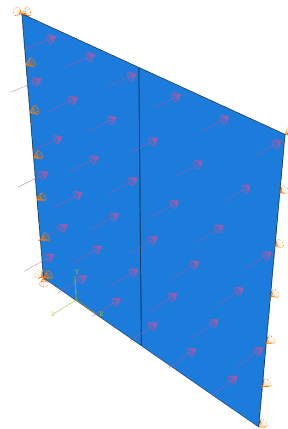


Figure 40: Boundary conditions

8.2.3 Load

The applied load was a distributed one P , applied on the whole top face of the plate, its value is reported in (Tab. 12).

8.2.4 Mesh

A 50 mm seed was selected to mesh the part, creating 1600 elements. The element type chose was a linear, full integration shell S4.

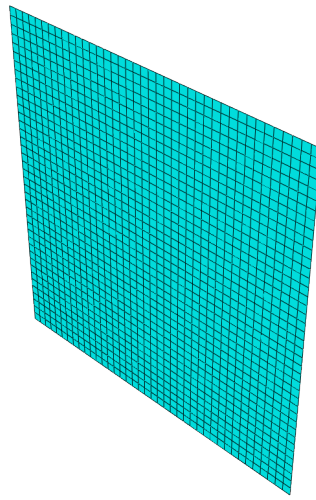


Figure 41: Meshed plate

8.3 Simulation results

In (Fig. 42) are presented the stresses and displacements acting on the plate. The vertical displacement in the centre of the plate is equal to:

$$\delta_{max} = 5.465 \text{ mm} \quad (8.1)$$

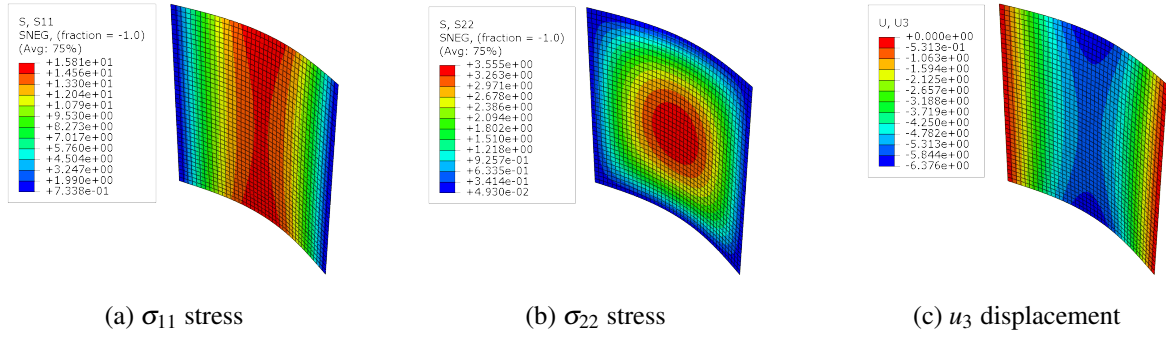


Figure 42: Stress and displacement acting on plate

The forces acting on the plate are presented in (Fig. 43).

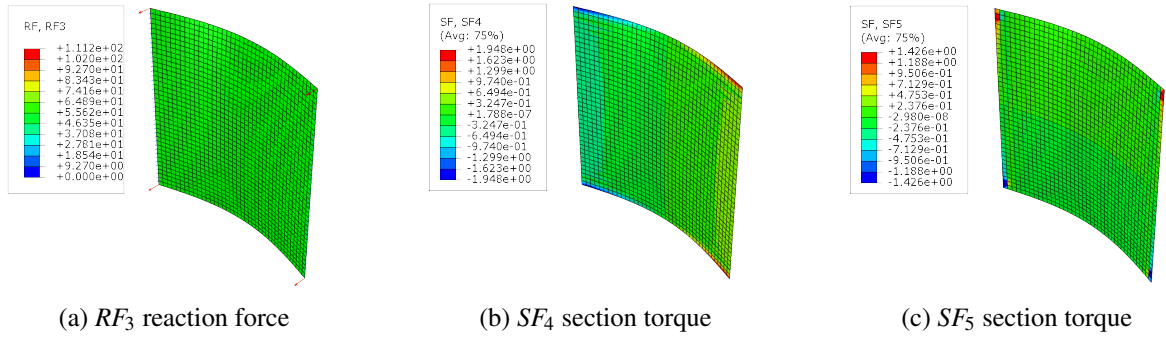


Figure 43: Internal and reaction forces acting on plate

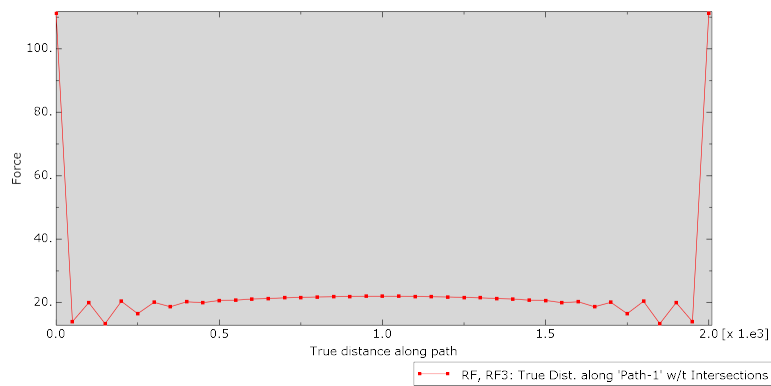


Figure 44: RF_3 plot along the edge

8.4 Result analysis

8.4.1 Analytical results

To verify and validate the analysis, a comparison against an analytical case was carried out. The whole plate was considered as a beam with length l , thickness t , with a distributed load $P_{beam} = P \cdot h = 1 \text{ N/mm}$ and rectangular cross section with $I = lt^3/12$.

In this model is possible to compute the maximum deflection in the beam centre with:

$$\delta_{max} = \frac{5pl^4}{384EI} = 6.695 \text{ mm} \quad (8.2)$$

This analytical value can be considered very close to the simulated one (Eq. 8.1) considering the hypothesis made and considering that the plate is being seen as a beam. Moreover, shell are inherently stiffer than beams, especially in the centre, since the load can be "distributed" to more material in both longitudinal and transversal directions, whereas, in a beam the load is opposed mostly by material along the longitudinal direction.

8.4.2 Comments on graphs

The vertical displacement u_3 presented in (Fig. 42c), shows that the displacement at the centre of the plate is smaller than the displacement at the centre of the free edges. This is due to the fibers at the centre having more material around them to counter the load and therefore deform less. Meanwhile the fibers at the free edges present material to resist the load only on one side, resulting in more vertical deformation.

Stress distribution inside the plate show similar results for σ_{11} in (Fig. 42a) and σ_{22} in (Fig. 42b): both have their maximum values in the centre of the beam, while at the edges stresses are null. The difference lays in the shape of the distributions: σ_{11} presents a distribution similar to u_3 , so with a central band presenting higher stresses and 0 at the supported edges; while σ_{22} presents null stresses along all edges, and a circular concentration in the centre, where it reaches its maximum value.

In (Fig. 43b) and (Fig. 43c) the section torques are presented: their combination is responsible for the typical lifting of the corners of the simply supported plate.

Regarding the vertical reaction forces on the edges, one should expect to have all the components equal to the load on the edge, divided by the number of elements, so $1000/40 = 25 \text{ N}$, but instead in (Fig. 43a) two bigger arrows are present on the corners, while along the rest of the edge the values are around 20 N . This behaviour is typical of shell elements, due to counter rotating torsional moments coming from two different faces, which can be seen as a force couple acting on the corners, effectively creating an additional concentrated reaction force.

RF3 values along the edge are plotted in (Fig. 44), where this typical behaviour of shells can be once again observed, the initial oscillations around the corners are due to numerical rounding, but if all the values are summed together the results is exactly the total load, so 1000 N .

9 Ex 9: Machine tool with shell elements

9.1 Scope of the exercise

This exercise's scope was to study again the machine tool presented in (Sec. 3.2.1), using shell elements instead of beam elements.

The structure was studied both statically, to extract the stresses and displacements, and dynamically, to extract its mode shapes and natural frequencies.

9.2 Model description

9.2.1 Geometry

The tool's geometry was the same presented in (Sec. 3.2.1). Instead of creating from scratch the structure, it was imported from a provided .cae file, where the tool's column and cross-head had already been modelled. Both of them already had a 8mm shell homogenous section with 5 integration points along the thickness assigned to them, and the cross-head already presented partitions where the load should be applied and where the columns should be positioned.

To properly recreate the structure, two columns and one cross-head were assembled together using a tie constraint. With this constraint, perfect contact between the parts is achieved, as they were one whole piece. Subsequently, using a view cut, a reference point was created in the centre of the cross-head's mid-plane to apply the load, then it was coupled with the whole mid-plane by selecting its node region. The material used is a classical steel, reported in (Tab. 13)

ρ	7860ton/mm ³
E	206000MPa
ν	0.33

Table 13: Material data

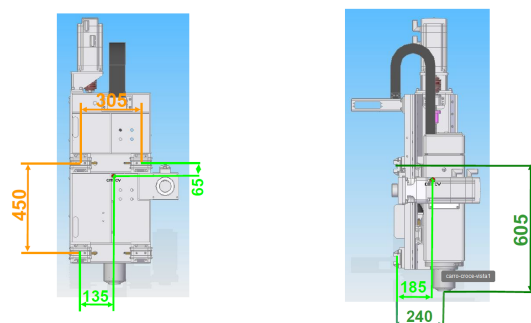


Figure 45: Toll-head centre of mass

9.2.2 Boundary conditions

The boundary conditions were applied to the bottom of the columns, constraining all the rigid displacements of the whole surface (Fig. 46).

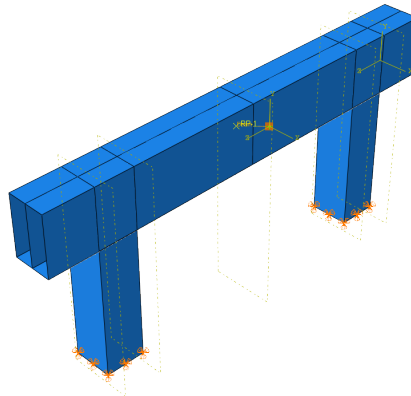


Figure 46: Boundary conditions

9.2.3 Load

The load applied was different for the two different analysis:

- In the static analysis, a vertical concentrated load $P = -500N$ directed downwards was applied, using a static general step, without Nlgeom.
- To define the deformed shapes of the structures a linear perturbation, frequency step was used, with 5 eigenvalues to be computed. Moreover, to correctly represent the problem, also the tool-head must be considered, with its mass and inertia, in its correct position, "outside" of the cross-head. An attachment point was created in the position of the head's centre of mass (Fig. 45), which was then connected to the side of the cross-head using wires. The wires were subsequently transformed into mechanical connections using beams, meaning that the connection could transfer both shear and torques to the cross-head.

Once the attachment point was correctly positioned, the inertia values (Tab. 14) of the tool head were inserted in Abaqus, having care of selecting the correct reference point centred in the centre of mass of the tool.

J_{11}	$627000ton/mm^2$
J_{22}	$68000ton/mm^2$
J_{33}	$669000ton/mm^2$

Table 14: Tool head's inertias

9.2.4 Mesh

The parts were already meshed inside the provided file resulting in (Fig. 47):

- Cross-head: 18512 S4R5 elements, which is a 4-node doubly curved thin shell, reduced integration, hourglass control, using five degrees of freedom per node.
- Columns: 3264 S4R5 elements, which is a 4-node doubly curved thin shell, reduced integration, hourglass control, using five degrees of freedom per node.

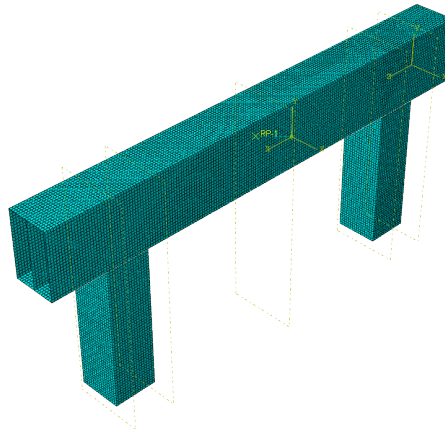


Figure 47: Mesh

9.3 Simulation results static

9.3.1 Displacements

In (Fig. 48) the structure's displacements are presented, the maximum displacement is equal to:

$$\delta_{max} = 1.392 \cdot 10^{-3} mm \quad (9.1)$$

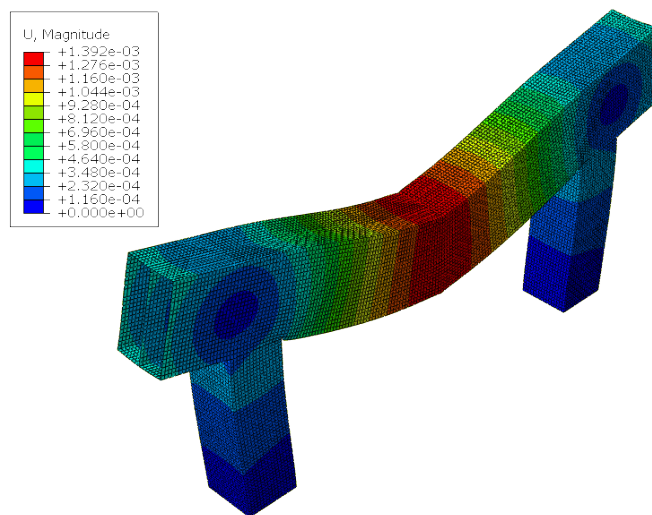


Figure 48: Displacements

9.3.2 Stresses

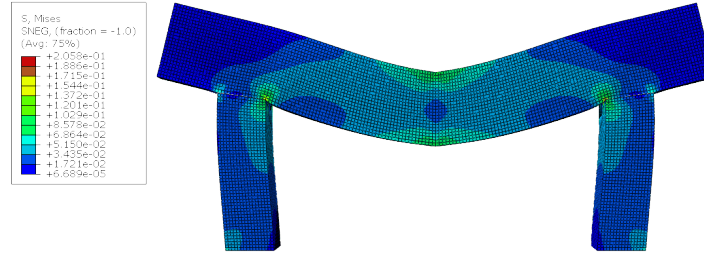


Figure 49: Von Mises stress

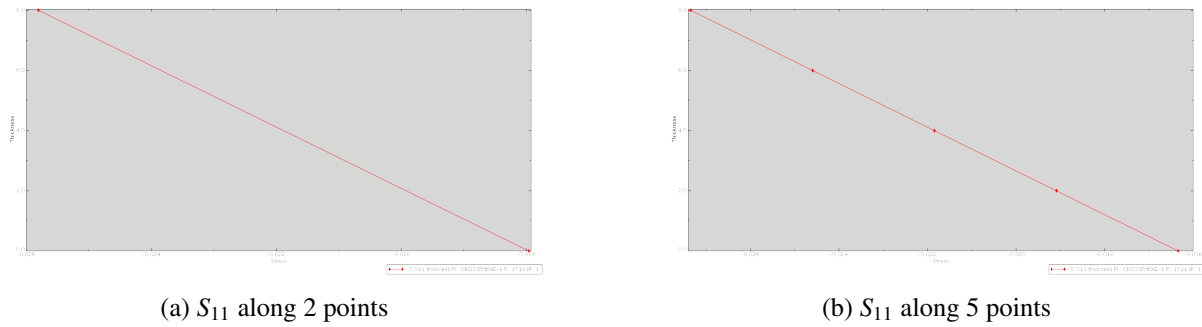


Figure 50: Stress distribution along shell thickness

9.4 Result analysis static

To verify the validity of the simulation, an analytical solution close to the studied case does not exist, therefore the results obtained in (Sec. 3.3.2) were compared. If two simulations of the same structure, with very different models and assumptions converge to the same result then it is safe to assume that the simulations are correct. The maximum displacement experienced with the two analysis indeed were very close considering the assumptions made in both, therefore the simulations can be regarded as sufficiently correct. Moreover, the shell elements result is closer to the one obtained with the shear-deformable one than the one obtained with the cubic formulation. This being due to the structure's beam is relatively thick, so results will be closer when using shells and shear-deformable beams.

δ_{max}^{shell}	$1.392 \cdot 10^{-3} mm$
δ_{max}^{cubic}	$1.205 \cdot 10^{-3} mm$
$\delta_{max}^{shear-def}$	$1.561 \cdot 10^{-3} mm$

Table 15: Maximum beam deflection

Analysing (Fig. 50) is possible to see a benefit of using shell elements. By default Abaqus shows the stress distribution along the shell with just the maximum and minimum value computed, without showing what happens between them. To see better stress development along the thickness it is advisable to request to the field output manager to show stresses values computed at all the integration points positioned along the thickness. In this way is actually possible to state that the stress distribution is linear in the shell. Beam elements may be faster, easier to setup, probably also more precise, but do not allow to see how stresses develop along the thickness, which, instead, is something that shell elements are capable of, thus deeming them more interesting for certain applications.

9.5 Simulation results dynamic

The first mode shape of the dynamic analysis, happening at $f_1 = 15.866\text{ Hz}$ is plotted in (Fig. 51).

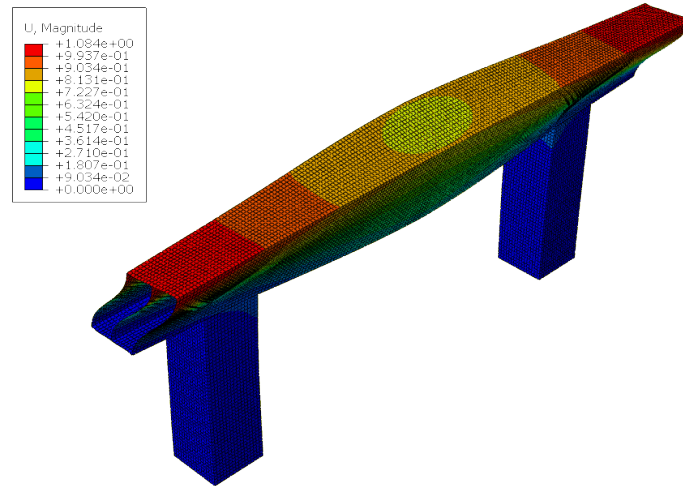


Figure 51: Mode shape at f_1

9.6 Result analysis dynamic

By looking at (Fig. 51), is reasonable to think the the results are correct, as one would expect the first mode shape to be some sort of bending towards the right, due to the head's position. Moreover, the cross-head section is weaker in lateral direction, explaining again why the structure deforms more laterally than in other directions.